Wilson Harper

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Houston, TX

Education

Rice University — Class of 2029

- B.S. Electrical & Computer Engineering; Minor in Entrepreneurship
- Recipient, Martel College Outstanding New Student Award (2026)
- Secretary, Martel College Parliament

Experience

Incoming Embedded Engineering Intern, RTX BBN Technologies — Summer 2026

- Selected to prototype embedded systems and robust PNT algorithms for harsh, GPS-denied environments

Undergraduate Researcher, Rice Experimental Pixels Lab — 2026 - Present

- Engineering a compact, highly directional wind system to synchronize physical airflow with XR environments
- Designing and integrating remote blowers, ducting, and control circuitry within strict spatial constraints
- Implementing UDP with networked POE-powered ESP32s, servos, and PWM fans

Avionics and Electrical Engineer, Rice Eclipse Rocketry — 2025 - Present

- Designing avionics architecture for an Active Flight Stabilization system intended for a spaceshot rocket
- Integrating RP2350 with I2C/UART/SPI sensors on a custom PCB to actuate servo canards in real-time
- Creating avionics bays with COTS components for high-reliability recovery systems

Teaching Assistant, BUSI 361: Communication for Entrepreneurs — 2026

- Facilitating instruction on communication, conflict resolution, networking, and workplace dynamics

Summer Camp Staff, Spokane, WA — 2025

- Responsible for up to 65 guests at a remote location accessible only by boat
- Handled high-stakes logistics, guest relations, and conflict resolution for attendees

Technical Projects

wilsonharper.net/projects

Custom LoRa Rocket Flight Computer & Telemetry Stack — 2026 - Present

- Designed, programmed, and integrated a custom RP2040-based flight computer in two weeks
- Engineered 1 Hz LoRa telemetry while logging altimeter, accelerometer, and GPS data at 20 Hz
- Built an accompanying ground station with custom antenna to display live data and Google Earth position

ADS-B Flight Tracking Station & Data Pipeline — 2025 - 2026

- Built Pi-powered ADS-B tracker featuring a custom-soldered 1090 MHz ground plane spider antenna
- Created secure data pipelines using Cloudflare Tunnels, Tailscale, and GitHub Pages

UAV with Position-Holding in GPS-Denied Environments — 2022 - 2025

- Led the design of a UAV capable of navigating GPS-denied environments with optical flow and LiDAR
- Implemented EKF sensor fusion, remote telemetry, modular airframe, and servo-actuated gripper

Doppler & Synthetic Aperture Radar at MIT Lincoln Laboratory RISE — 2024

- One of ~4% of applicants selected for DoD-sponsored free research experience
- Researched, built, and programmed Doppler and synthetic aperture radars, mentored by Lincoln Labs staff

Skills

- **Certifications:** Technician Class Amateur Radio License, Tripoli Level 1 High Powered Rocketry Certification
- **Hardware & Systems:** LiDAR, radar, sensor integration, UAVs, PCB design, microcontrollers, LoRa, telemetry
- **Software:** Python, C/C++, CAD (Fusion, Onshape), ArduPilot, Cloudflare, Tailscale, KiCad
- **Tools:** oscilloscopes, soldering, 3D printing, CNC routing, hand tools, multimeters, breadboards